an SPE that has a value *worse* than that associated with repetition of the one-period Nash outcome. Following APS, we shall soon see that the best and worst equilibrium values are linked.

25.8. Values of all SPEs

The role played by the lemma in analyzing our two examples hints at the central role that it plays in methods that APS developed for describing and computing values for *all* the subgame perfect equilibria.[12] APS build on the way that the lemma characterizes SPE values in terms of a first-period competitive equilibrium outcome, along with a pair of continuation values, each element of which is itself a value associated with some SPE. The lemma directs APS's attention away from a *set of strategy profiles* σ and toward a *set of values* $V_g(\sigma)$ associated with those profiles. They define the set V of values associated with subgame perfect equilibria:

$$V = \{V_g(\sigma) \mid \sigma \text{ is an SPE}\}.$$

Evidently, $V \subset I\!R$.

Lemma 2: A competitive equilibrium $(x, y) \in C$ is supported by an SPE, i.e. $x = \sigma_1^h$ and $y = \sigma_1^g$, if and only if there exists a function $v(\cdot) : Y \to V$ such that

$$(1-\delta)\, r(x,y) + \delta v(y) \;\geq\; (1-\delta)\, r(x,\eta) + \delta v(\eta), \qquad \forall\, \eta \in Y. \tag{25.8.1}$$

Proof: *Necessity.* Suppose $(x, y) \in C$ is supported by an SPE σ, so that $x = \sigma_1^h$ and $y = \sigma_1^g$. By condition (1) of Lemma 1, for each $\eta \in Y$ the continuation strategy $\sigma|_{(x,\eta)}$ is itself an SPE. Define $v(\eta) = V_g(\sigma|_{(x,\eta)}) \in V$ for each $\eta \in Y$. By condition (3) of Lemma 1, we then have

$$(1-\delta)\, r(x,y) + \delta V_g(\sigma|_{(x,y)}) \;\geq\; (1-\delta)\, r(x,\eta) + \delta V_g(\sigma|_{(x,\eta)}),$$

which is exactly (25.8.1) with $v(\eta) = V_g(\sigma|_{(x,\eta)})$.

[12] Relative to earlier editions, this section has been expanded and revised to take account of contributions by Barthélemy and Mengus (2020).

Sufficiency. Suppose there exists $v(\cdot) : Y \to V$ satisfying (2). Since $v(\eta) \in V$ for each $\eta \in Y$, there exists for each η an SPE $\tilde{\sigma}(\eta)$ with $V_g(\tilde{\sigma}(\eta)) = v(\eta)$. Construct a strategy profile σ by setting $\sigma|_{(x,\eta)} = \tilde{\sigma}(\eta)$ for all $\eta \in Y$, and (without loss of generality) $\sigma|_{(\nu,\eta)} = \tilde{\sigma}(\eta)$ for all $\nu \in X$. By construction this profile satisfies all three conditions of Lemma 1 and is therefore an SPE with $\sigma_1^h = x$ and $\sigma_1^g = y$. ∎

Definition 7: Let $W \subset \mathbf{R}$. A triple $(x, y, w(\cdot))$ is said to be *admissible with respect to* W if $(x, y) \in C$ and $w(\cdot) : Y \to W$ is a function satisfying

$$(1-\delta)\, r(x,y) + \delta w(y) \;\geq\; (1-\delta)\, r(x,\eta) + \delta w(\eta), \qquad \forall\, \eta \in Y. \tag{25.8.2}$$

Notice that when $W \subset V$, the admissible 4-tuple (x, y, w_1, w_2) determines an SPE with strategy profile

$$\sigma_1 = (x, y),\ \sigma|_{(x,y)} = \sigma^1,\ \sigma|_{(\nu,\eta)} = \sigma^2 \ \text{ for } (\nu,\eta) \neq (x,y)$$

where σ_1 is a continuation strategy that yields value $w_1 = V_g(\sigma^1)$ and σ_2 is a strategy that yields continuation value $w_2 = V_g(\sigma^2)$. The value of the SPE is $V_g(\sigma) = w = (1-\delta)\, r(x,y) + \delta w_1$.

25.8.1. Basic idea of dynamic programming squared

We want to find the set V. In Definition 7, W serves as a set of candidate continuation values. The idea is to pick an $(x, y) \in C$, then to check whether we can find $(w_1, w_2) \in W \times W$ that would make the government want to adhere to the y component if w_1 and w_2 could be used as continuation values for adhering to and deviating from y, respectively. If the answer is yes, we say that the 4-tuple (x, y, w_1, w_2) is "admissible with respect to W". Because we have verified that the incentive constraints are satisfied, a yes answer allows us to calculate the *value* (i.e., the left side of (25.8.2)) that can be supported with w_1, w_2 as continuation values. Thus, the idea is to use (25.8.2) to define a mapping from values tomorrow to values today, like that used in dynamic programming. In the next section, we'll define $B(W)$ as the set of possible values attained with admissible pairs of continuation values drawn from $W \times W$. Then we'll view B as an operator mapping sets of continuation values W into sets of values

$B(W)$. This operator is the counterpart to the T operator associated with ordinary dynamic programming.

To pursue this analogy, recall the Bellman equation associated with the McCall model of chapter 7:

$$Q = \int \max\left\{\frac{w}{1-\beta}, c + \beta Q\right\} d\,F(w).$$

Here $Q \in I\!R$ is the expected discounted value of an unemployed worker's income *before* he has drawn a wage offer. The right side defines an operator $T(Q)$, so that the Bellman equation is

$$Q = T(Q). \tag{25.8.3}$$

Just as the right side of (25.8.3) takes a candidate continuation value Q for tomorrow and maps it into a value $T(Q)$ for today, APS define a mapping $B(W)$ that, by considering only admissible 4-tuples, maps a *set* of values W tomorrow into a new *set* $B(W)$ of values today. Thus, APS use admissible 4-tuples to map candidate continuation values tomorrow into new candidate values today. In the next section, we'll iterate to convergence on $B(W)$, but as we'll see, it won't work to start from just any initial set W. We have to start from a big enough set.

25.9. APS machinery

Definition 8: For each set $W \subset \mathbf{R}$, let $B(W)$ be the set of possible values

$$w \;=\; (1-\delta)\, r(x,y) + \delta\, w(y)$$

associated with admissible triples $(x, y, w(\cdot))$ with respect to W.

Think of W as a set of potential continuation values and $B(W)$ as the set of values that they support. From the definition of admissibility it immediately follows that the operator B is *monotone*.

Property (monotonicity of B): If $W \subseteq W' \subseteq R$, then $B(W) \subseteq B(W')$.

Proof: Suppose $w \in B(W)$. Then there exists an admissible triple $(x, y, w(\cdot))$ with respect to W such that $w = (1-\delta)r(x,y) + \delta w(y)$. Since $W \subseteq W'$, the

function $w(\cdot)$ maps Y into W' as well, so $(x, y, w(\cdot))$ is admissible with respect to W' and $w \in B(W')$. ∎

It can also be verified that $B(\cdot)$ maps compact sets W into compact sets $B(W)$.

The self-supporting character of subgame perfect equilibria is referred to in the following definition:

DEFINITION 9: The set W is said to be *self-generating* if $W \subseteq B(W)$.

Thus, a set of continuation values W is said to be self-generating if it is contained in the set of *values* $B(W)$ that are generated by pairs of continuation values selected from W. This description makes us suspect that if a set of values is self-generating, it must be a set of SPE values. Indeed, notice that by virtue of the lemma, the set V of SPE values $V_g(\sigma)$ is self-generating. Thus, we can write $V \subseteq B(V)$. APS show that V is the *largest* self-generating set. The key to showing this point is the following theorem:[13]

THEOREM 1 (A self-generating set is a subset of V): If $W \subset I\!R$ is bounded and self-generating, then $B(W) \subseteq V$.

The proof is based on "forward induction" and proceeds by taking a point $w \in B(W)$ and constructing an SPE with value w.

PROOF: Assume $W \subseteq B(W)$. Choose $w \in B(W)$ and construct an SPE with value w as follows.

Step 1. Because $w \in B(W)$, there exist $(x_1, y_1) \in C$ and a function $w_1(\cdot) : Y \to W$ such that

$$w \;=\; (1-\delta)\, r(x_1, y_1) + \delta\, w_1(y_1) \;\geq\; (1-\delta)\, r(x_1, \eta) + \delta\, w_1(\eta), \quad \forall\, \eta \in Y.$$

Set $\sigma_1 = (x_1, y_1)$.

Step 2. Since $w_1(y_1) \in W \subseteq B(W)$, there exist $(x_2, y_2) \in C$ and a function $w_2(\cdot) : Y \to W$ such that

$$w_1(y_1) \;=\; (1-\delta)\, r(x_2, y_2) + \delta\, w_2(y_2) \;\geq\; (1-\delta)\, r(x_2, \eta) + \delta\, w_2(\eta), \quad \forall\, \eta \in Y.$$

[13] The *unbounded* set $I\!R$ (the extended real line) is self-generating but not meaningful. It is self-generating because any value $v \in I\!R$ can be supported if there are no limits on the continuation values. It is not meaningful because most points in $I\!R$ are values that cannot be attained with *any* strategy profile.

Set the first-period outcome of the continuation economy (occurring given that y_1 was chosen in period 1) equal to (x_2, y_2); that is, set $(\sigma|_{(x_1,y_1)})_1 = (x_2, y_2)$.

Continuation. Iterating in this way, for each $w \in B(W)$ we construct sequences of functions $w_1(\cdot), w_2(\cdot), w_3(\cdot), \ldots$ and first-period outcomes $(x_1, y_1), (x_2, y_2), (x_3, y_3), \ldots$. Since $\delta < 1$ and W is bounded,

$$w \;=\; \lim_{t\to\infty}\left[(1-\delta)\sum_{s=1}^{t}\delta^{s-1}r(x_s, y_s) + \delta^t w_t(y_t)\right] \;=\; (1-\delta)\sum_{s=1}^{\infty}\delta^{s-1}r(x_s, y_s).$$

At each stage the continuation values make one-period deviations unimprovable, so the constructed strategy profile is an SPE. By construction $V_g(\sigma) = w$, so $w \in V$. ∎

Compactness of V follows by an argument parallel to the proof of Theorem 4 in Abreu, Pearce, and Stacchetti (1990).

PROPOSITION 2: The set V is compact.

PROOF: We know that V is bounded and self-generating ($V \subseteq B(V)$). Let $\mathrm{cl}(V)$ denote the closure of V. Since V is bounded, $\mathrm{cl}(V)$ is compact and $V \subseteq \mathrm{cl}(V)$.

By monotonicity of B,

$$V \;=\; B(V) \;\subseteq\; B(\mathrm{cl}(V)).$$

Since $B(\cdot)$ maps compact sets into compact sets, $B(\mathrm{cl}(V))$ is compact. Because $\mathrm{cl}(V)$ is the smallest closed set containing V, and $V \subseteq B(\mathrm{cl}(V))$, it follows that

$$\mathrm{cl}(V) \;\subseteq\; B(\mathrm{cl}(V)),$$

so $\mathrm{cl}(V)$ is bounded and self-generating. Theorem 1 then gives $\mathrm{cl}(V) \subseteq V$, and therefore $V = \mathrm{cl}(V)$, which means V is compact. ∎

COROLLARY: (Two-value characterization.) A competitive equilibrium $(x, y) \in C$ is supported by an SPE, that is $x = \sigma_1^h$ and $y = \sigma_1^g$, if and only if there exist $(v_1, v_2) \in V \times V$ such that

$$(1-\delta)\, r(x, y) + \delta v_1 \;\geq\; (1-\delta)\, r(x, \eta) + \delta v_2, \qquad \forall\, \eta \in Y.$$

PROOF: *Sufficiency* is immediate: given $(v_1, v_2) \in V \times V$, define $v(y) = v_1$ and $v(\eta) = v_2$ for $\eta \neq y$; then (25.8.1) holds, so Lemma 2 applies.

Necessity. Since (x, y) is supported by an SPE, Lemma 2 yields a function $v(\cdot) : Y \to V$ satisfying (25.8.1). Set $v_1 = v(y)$. The set $\{v(\eta) : \eta \in Y\}$ is a subset of V; since V is compact and bounded, its infimum is attained in V. Set

$$v_2 \;=\; \inf_{\eta \in Y} v(\eta) \;\in\; V.$$

Then $v(\eta) \geq v_2$ for all $\eta \in Y$, so (25.8.1) implies

$$(1-\delta)\, r(x,y)+\delta v_1 \;=\; (1-\delta)\, r(x,y)+\delta v(y) \;\geq\; (1-\delta)\, r(x,\eta)+\delta v(\eta) \;\geq\; (1-\delta)\, r(x,\eta)+\delta v_2,$$

for all $\eta \in Y$, as required. ∎

Collecting results, we know that

1. $V \subseteq B(V)$ (by the lemma).
2. If $W \subseteq B(W)$, then $B(W) \subseteq V$ (by theorem 1).
3. B is monotone and maps compact sets into compact sets.

Facts 1 and 2 imply that $V = B(V)$, so that the set of equilibrium values is a "fixed point" of B, in particular, the *largest* bounded fixed point. Monotonicity of B and the fact that it maps compact sets into compact sets provides an algorithm for computing the set V, namely, to start with a set W_0 for which $V \subseteq B(W_0) \subseteq W_0$, and to iterate on B. In more detail, we use the following steps:

1. Start with a set $W_0 = [\underline{w}_0, \overline{w}_0]$ that we know is bigger than V, and for which $B(W_0) \subseteq W_0$. It will always work to set $\overline{w}_0 = \max_{(x,y)\in C} r(x,y)$, $\underline{w}_0 = \min_{(x,y)\in C} r(x,y)$.
2. Compute the boundaries of the set $B(W_0) = [\underline{w}_1, \overline{w}_1]$. The value $\overline{w}_1$ solves the problem
$$\overline{w}_1 = \max_{(x,y)\in C} (1-\delta)\, r(x,y) + \delta \overline{w}_0$$
subject to
$$(1-\delta)\, r(x,y) + \delta \overline{w}_0 \geq (1-\delta)\, r(x,\eta) + \delta \underline{w}_0 \quad \text{for all} \quad \eta \in Y.$$
The value $\underline{w}_1$ solves the problem
$$\underline{w}_1 = \min_{(x,y)\in C;\, (w_1,w_2)\in[\underline{w}_0,\overline{w}_0]^2} (1-\delta)\, r(x,y) + \delta w_1$$

subject to

$$(1-\delta)\, r(x,y) + \delta w_1 \geq (1-\delta)\, r(x,\eta) + \delta w_2 \quad \forall\, \eta \in Y.$$

With $(\underline{w}_0, \overline{w}_0)$ chosen as before, it will be true that $B(W_0) \subseteq W_0$.

3. Having constructed $W_1 = B(W_0) \subseteq W_0$, continue to iterate, producing a decreasing sequence of compact sets $W_{j+1} = B(W_j) \subseteq W_j$. Iterate until the sets converge.

In section 25.13, we will present a direct way to compute the best and worst SPE values, one that evades having to iterate on the B operator.

25.10. Self-enforcing SPE

A subgame perfect equilibrium with a *worst* value $v \in V$ has the remarkable property that it is "self-enforcing." We use the following definition:

Definition 10: A subgame perfect equilibrium σ with first-period outcome $(\tilde{x}, \tilde{y}) \in C$ is said to be *self-enforcing* if

$$\sigma|_{(x,y)} = \sigma \qquad \text{if } \ (x,y) \neq (\tilde{x}, \tilde{y}). \tag{25.10.1}$$

A strategy profile satisfying equation (25.10.1) is called self-enforcing because after a one-shot deviation the consequence is simply to restart the equilibrium.

Recall our earlier characterization of a competitive equilibrium as a pair $(h(y), y)$, where $x = h(y)$ is the mapping from the government's action y to the private sector's equilibrium response. The value $\underline{v}$ associated with the worst subgame perfect equilibrium σ satisfies

$$\underline{v} = \min_{y,v} \left\{ (1-\delta)\, r(h(y), y) + \delta v \right\} = (1-\delta) r(h(\tilde{y}), \tilde{y}) + \delta \tilde{v}, \tag{25.10.2}$$

where the minimization is subject to $y \in Y$, $v \in V$, and the incentive constraint

$$(1-\delta)\, r(h(y), y) + \delta v \geq (1-\delta)\, r(h(y), \eta) + \delta \underline{v} \quad \text{for all } \ \eta \in Y. \tag{25.10.3}$$

Let $\tilde{v}$ be a continuation value that attains the right side of equation (25.10.2), and let $\sigma_{\tilde{v}}$ be a subgame perfect equilibrium that supports continuation value

$\tilde{v}$. Let $(\tilde{x}, \tilde{y})$ be the first-period outcome that attains the right side of equation (25.10.2). Thus, $\underline{v} = (1-\delta)r(\tilde{x}, \tilde{y}) + \delta\tilde{v}$. Since $\underline{v}$ is both the continuation value when first-period outcome $(x, y) \neq (\tilde{x}, \tilde{y})$ *and* the value associated with subgame perfect equilibrium σ, it follows that

$$\begin{aligned} \sigma_1 &= (\tilde{x}, \tilde{y}) \\ \sigma|_{(x,y)} &= \begin{cases} \sigma_{\tilde{v}} & \text{if } (x, y) = (\tilde{x}, \tilde{y}) \\ \sigma & \text{if } (x, y) \neq (\tilde{x}, \tilde{y}). \end{cases} \end{aligned} \tag{25.10.4}$$

Because of the double role played by $\underline{v}$, i.e., $\underline{v}$ is both the value of equilibrium σ and the "punishment" continuation value of the right side of the incentive constraint (25.10.3), an equilibrium strategy σ that supports $\underline{v}$ is self-enforcing.[14]

The preceding argument thus establishes

Proposition 2: A subgame perfect equilibrium σ associated with $\underline{v} = \min\{v : v \in V\}$ is self-enforcing.

25.10.1. The quest for something worse than repetition of Nash outcome

Notice that the first subgame perfect equilibrium that we computed, whose outcome was infinite repetition of the one-period Nash equilibrium, is a self-enforcing equilibrium. However, in general, the infinite repetition of the one-period Nash equilibrium is not the *worst* subgame perfect equilibrium. This fact opens the possibility that even when reversion to Nash after a deviation is *not* able to support repetition of Ramsey as an SPE, we might still support repetition of the Ramsey outcome by reverting to a SPE with a value worse than that associated with repetition of the Nash outcome whenever the government deviates from an expected one-period choice.

[14] As we show below, the structure of the programming problem, with the double role played by $\underline{v}$, makes it possible to compute the worst value directly.